

Today's Content

- Trailing Zeros in factorial
- Properties of %
- Power function
- fast power function
- Fermat's Theorem
- Pair Sum Divisible by M

factorial

$$\text{fact}(5) = 1 * 2 * 3 * 4 * 5 = 120 \rightarrow 1$$

Ques! Given an integer N , Count the no. of trailing 0's in $N!$

$N \rightarrow$	1	2	3	4	5	6	7	8	9	10
$\text{fact} \rightarrow$	1	2	6	24	120	720	5040	40320	362880	3628800
trailing 0's $\rightarrow$	0	0	0	0	1	1	1	1	1	2

$\text{fact}(n) \rightarrow$ can be very large $\rightarrow$ can't be stored in int/long.

$$\text{fact}(10) = 1 * 2 * 3 * 4 * 5 * 6 * 7 * 8 * 9 * 10$$

$\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$
 2 2*2 5 2 2*2*2 2*5

Ans = # of times 5 is multiplied

$$\text{fact}(30) = 1 * \dots * 5 * \dots * 10 * \dots * 15 * \dots * 20 * \dots * 25 * \dots * 30$$

$\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$
 1 1 1 1 2 1

Ans = 7

Ques What is the count of multiples of x from 1 to N ?

$$x = 3$$

$$N = 10$$

1 2 3 4 5 6 7 8 9 10

$$\text{Ans} = N/x$$

$$10/3 = 3$$

$$x = 5, N = 100$$

$$\frac{100}{5} = 20$$

$$\text{fact}(30) = 1 * \dots * 5 * \dots 10 * \dots 15 * \dots 20 * \dots 25 * \dots 30$$

Ans = 7

$\downarrow$
 $5 * 5$

$$\begin{aligned} \# \text{ multiples of } 5 \text{ from } 1 \text{ to } 30 &= 30/5 = 6 \\ \# \text{ multiples of } 25 \text{ from } 1 \text{ to } 30 &= 30/25 = 1 \\ &\underline{7} \end{aligned}$$

$$N=100 \rightarrow \text{Ans} = 24$$

$$\# \text{ multiples of } 5 \text{ from } 1 \text{ to } 100 = 100/5 = 20$$

$$\# \text{ multiples of } 25 \text{ from } 1 \text{ to } 100 = 100/25 = 4$$

24

$$\{25, 50, 75, 100\}$$

$\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$
 $5*5$ $5*5*2$ $5*5*3$ $5*5*4$

$$N=300 \rightarrow \text{Ans} = ?$$

$$\# \text{ multiples of } 5 \text{ from } 1 \text{ to } 300 = 300/5 = 60$$

$$\# \text{ multiples of } 25 \text{ from } 1 \text{ to } 300 = 300/25 = 12$$

72

$$\begin{aligned} 125 &\rightarrow 5 * 5 * 5 \\ 250 &\rightarrow 5 * 5 * 5 * 2 \end{aligned}$$

$$\# \text{ multiples of } 125 \text{ from } 1 \text{ to } 300 = 300/125 = 2$$

74 $\rightarrow$ Answer

$$\begin{aligned} \text{Ans} = & \# \text{ multiples of } 5 + \\ & \# \text{ multiples of } 5^2 + \\ & \# \text{ multiples of } 5^3 + \\ & \vdots \end{aligned}$$

$$\# \text{ multiples of } 5^K \text{ in range } 1 \text{ to } N$$

$$K = \log_5 N$$

$\downarrow$
maximum value of K

$$5^K \leq N$$

Pseudo code

ans = 0

for (x = 5; x ≤ N; x = x * 5)

 ans = ans + (N/x)

}

return ans;

T.C = $O(\log_5(N))$

S.C = $O(1)$

Todo:- Try out with Gf Approach. T.C = $O(\log N)$ as we used to calculate power

Modular Arithmetic

$\% \rightarrow ? \rightarrow$ Gives you the remainder when 2 integers are divided.

$a \% b$ = remainder when a is divided by b .

$$99 \% 8 = 3$$

$$\text{Dividend} = \text{Divisor} \times \text{Quotient} + \text{Remainder}$$

$$\text{Remainder} = \text{Dividend} - (\text{Divisor} \times \text{Quotient})$$

$\downarrow$
Max multiple of divisor $\leq$ dividend

$$\begin{aligned} &= 99 - 96(8 \times 12) \\ &= 3 \end{aligned}$$

$$\begin{aligned} \Rightarrow -40 \% 7 &= -40 - (\text{max multiple of } 7 \leq -40) \\ &= -40 - (-42) = 2 \end{aligned}$$

Note: remainder is always +ve

$$\begin{aligned} \Rightarrow -60 \% 9 &= -60 - (\text{max multiple of } 9 \leq -60) \\ &= -60 - (-63) \\ &= 3 \end{aligned}$$

C/C++/Java --

vs

Python

$$-70 \% 7 = -5 + 7 = 2$$

2

$$-60 \% 9 = -6 + 9 = 3$$

3

```
if (dividend < 0) {  
    ans = (dividend % divisor) + divisor  
}
```

Why we are studying **Modulus**? $\rightarrow$ Helps us limit values in a range.

$$\left. \begin{array}{c} -\infty \\ \vdots \\ +\infty \end{array} \right\} \% 10 \Rightarrow \left\{ \begin{array}{c} 0 \\ \vdots \\ 9 \end{array} \right.$$

$\%$ 10 limits values in the range $[0, 9]$

Quiz Max. value of $a \% M$?

$$a \rightarrow \% M \rightarrow \left\{ \begin{array}{c} 0 \\ \vdots \\ M-1 \end{array} \right.$$

min val. $\{$ when a is a multiple of M $\}$
 $M-1 \rightarrow$ max value

Properties.

$$1) (a+b) \% m = ((a \% m + b \% m) \% m)$$

e.g. $a=4; b=5; m=6$

$$(4+5) \% 6$$

$$9 \% 6$$

$$= 3$$

$$4 \% 6 + 5 \% 6$$

$$4 + 5 = 9 \% 6 = 3$$

$$\underbrace{a \% m}_{[0, m-1]} + \underbrace{b \% m}_{[0, m-1]} \rightarrow \text{max of this can exceed } m-1$$

$$2) (a \% m) = (a+m) \% m$$

$$\text{RHS} \Rightarrow (a+m) \% m$$

$$\Rightarrow (a \% m + \cancel{m \% m}) \% m$$

$$\Rightarrow (a \% m) \% m$$

$$\Rightarrow (a \% m)$$

$$3) (a * b) \% m = (a \% m * b \% m) \% m$$

$$4) (a-b) \% m = (a \% m - b \% m + m) \% m$$

e.g. $a=8, b=4, m=5$

$$(8-4) \% 5$$

$$4 \% 5 = 4$$

$$(8 \% 5 - 4 \% 5) \% 5$$

$$(3 - 4) \% 5 = (-1) \% 5 = -1$$

$$(a-b) \% m = (\underbrace{a \% m}_{[0, m-1]} - \underbrace{b \% m}_{[0, m-1]} + m) \% m$$

$$\begin{aligned} \text{min value} &= 0 - (m-1) \\ &= 1-m \text{ (this can be -ve)} \end{aligned}$$

$$(a-b) \% m = (a \% m - b \% m + m) \% m$$

→ Holds true for all languages.

Questions after break.

10:45 PM

Q Given a and n . Calculate a^n

$$\begin{aligned} a^{10} &= a^5 * a^5 \\ a^{12} &= a^6 * a^6 \\ a^{13} &= a * a^6 * a^6 \\ a^{15} &= a * a^7 * a^7 \end{aligned}$$

idea:- a^n :-
if $(N \% 2 == 0)$
 $a^{N/2} * a^{N/2}$
else
 $a^{N/2} * a^{N/2} * a$

```
power(a, n) {  
    if (N % 2 == 0)  
        power(a, n/2) * power(a, n/2);  
    else  
        power(a, n/2) * power(a, n/2) * a;  
}
```

FAST EXPONENTIATION

$$N = 10 \rightarrow 5 \rightarrow 2 \rightarrow 1 \rightarrow 0$$

```
power(a, n) {  
    if (n == 0) return 1;  
    h = power(a, n/2)  
    ha = h * h;  
    if (N % 2 == 0) {  
        return ha;  
    }  
    else {  
        return ha * a;  
    }  
}
```

$$T.C = O(\log_2 N)$$

$a^n \% p$

T.C = $O(\log_2 N)$
S.C = $O(1)$

```
power(a, n, p) {
    if (n == 0) return 1;
    long he = power(a, n/2, p);
    long ha = (he * p + he * p) % p;
    if (n % 2 == 0) {
        return ha;
    }
    else {
        return (ha * p + a * p) % p;
    }
}
```

$$1 \leq (a, N, p) \leq 10^9$$

$$\max p = 10^9$$

$$((he \% p) * (he \% p)) \% p$$

$$\begin{matrix} \downarrow & \downarrow \\ [0, p-1] & [0, p-1] \\ \max = p-1 & \max = p-1 \end{matrix}$$

$$(p-1) * (p-1) \approx p^2 \approx 10^{18}$$

$$* (a|b) \% m = (a * b^{-1}) \% m = (a \% m * b^{-1} \% m) \% m$$

→ inverse mod of b wrt m

$$10/5 = 2$$

$$10/4 = 2.5$$

$$20.5/3 = X$$

$$\frac{b}{b} \% m \rightarrow (b * b^{-1}) \% m$$

↓
1

e.g. $5^{-1} \bmod 7$
 $= (5 * 5^{-1}) \% 7 = 1$

$$(5 * 1) \% 7 = 5$$

$$(5 * 2) \% 7 = 3$$

$$(5 * 3) \% 7 = 1$$

$$5^{-1} \bmod 7 = 3$$

e.g. $3^{-1} \bmod 13$
 $= (3 * ?) \% 13 = 1$

$$(3 * 1) \% 13 = 3$$

$$(3 * 9) \% 13 = 1$$

$$(3 * 2) \% 13 = 6$$

$$(3 * 3) \% 13 = 9$$

$$(3 * 4) \% 13 = 12$$

$$3^{-1} \bmod 13 = 9$$

$$(x * \text{inv}(x)) \% m = 1$$

↳ Mathematical Expression

$$x^{-1} \bmod (m) \rightarrow (x * ?) \% m = 1$$

↓
Range

$$0 \leq \quad \leq m-1$$

→ only possible to calculate if

$$\gcd(x, m) = 1$$

→ why?
we won't discuss.

Brute force:- $T.C = O(m)$

Linear
Diophantine
Equation

Fermat Theorem

$$a^{p-1} \equiv 1 \pmod{p}$$

→ prime number 10^9+7

$$x \equiv y \pmod{m}$$

$$\Rightarrow x \% m = y \% m$$

$$\Rightarrow a^{p-1} \% p = 1 \% p$$

$$\Rightarrow a^{p-1} * a^{-1} \% p = 1 * a^{-1} \% p$$

$$\Rightarrow a^{p-2} \% p = a^{-1} \% p \quad \rightarrow \text{Inverse Modulo}$$

→ Fast Power Exponentiation

$$T.C = O(\log p)$$

Condition:- p is prime number.

Q find $3^{1002} \% 11$?

* 11 is a prime no.

$$3^{11-1} \% 11 = 3^{10} \% 11 = 1$$

$$a^{p-1} \% p = 1 \% p = 1$$

$$3^{1002} = \left(\underbrace{3^{10} * 3^{10} * 3^{10} \dots * 3^{10}}_{100 \text{ times}} * 3^2 \right) \% 11$$

$$= \left((3^{10} \% 11 * 3^{10} \% 11 * \dots * 3^{10} \% 11) * (3^2 \% 11) \right) \% 11$$

$$= 9 \% 11 = 9$$